Assignment – Kubernetes Deployment

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1. Objective

This assignment demonstrates deploying and managing an application in Kubernetes.
Key concepts shown:

- Pods
- ReplicaSets
- Deployments with rolling updates
- Probes (liveness, readiness, startup)
- Service exposure
- Version updates (v1 → v2)

2. Application Overview

- **Frontend (mandatory)**: Simple HTML/CSS/JS app
  - v1 → Displays *Frontend Application – Version 1* (blue theme)
  - v2 → Displays *Frontend Application – Version 2* (pink theme)
- **Backend (optional)**: Any API framework (not required for demo)

3. Docker Images

-> Dockerfile (Unified)

```
FROM nginx:alpine
COPY index.html /usr/share/nginx/html/index.html
```

-> Build & Push

```
docker build -t rpoojitha04/frontend:v1 .
docker push rpoojitha04/frontend:v1
docker build -t rpoojitha04/frontend:v2 .
docker push rpoojitha04/frontend:v2
```

4. Kubernetes YAML Files

Pod

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: frontend-pod
  labels:
    app: frontend
spec:
  containers:
    - name: frontend
      image: rpoojitha04/frontend:v1
      ports:
```

```
      - containerPort: 80
```

ReplicaSet

```
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: frontend-rs
  labels:
    app: frontend
spec:
  replicas: 3
  selector:
    matchLabels:
      app: frontend
  template:
    metadata:
      labels:
        app: frontend
    spec:
      containers:
        - name: frontend
          image: rpoojitha04/frontend:v1
          ports:
            - containerPort: 80
```

Deployment

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: frontend-deployment
  labels:
    app: frontend
spec:
  replicas: 3
  selector:
    matchLabels:
      app: frontend
  strategy:
    type: RollingUpdate
    rollingUpdate:
      maxUnavailable: 1
      maxSurge: 1
  template:
    metadata:
```

```
    labels:
      app: frontend
  spec:
    containers:
      - name: frontend
        image: rpoojitha04/frontend:v1
        ports:
          - containerPort: 80
        livenessProbe:
          httpGet:
            path: /
            port: 80
          initialDelaySeconds: 5
          periodSeconds: 10
        readinessProbe:
          httpGet:
            path: /
            port: 80
          initialDelaySeconds: 5
          periodSeconds: 10
        startupProbe:
          httpGet:
            path: /
            port: 80
          failureThreshold: 30
          periodSeconds: 10
```

Service

```
apiVersion: v1
kind: Service
metadata:
  name: frontend-service
spec:
  type: NodePort
  selector:
    app: frontend
  ports:
    - port: 80
      targetPort: 80
      nodePort: 30007
```

5. Deployment Steps

kubectl apply -f pod.yml

kubectl apply -f rs.yml
kubectl apply -f deployment.yml
kubectl apply -f service.yml

```
PS C:\Users\hp\Documents\GenAI\Assignment2> kubectl apply -f pod.yaml
pod/frontend-pod created
PS C:\Users\hp\Documents\GenAI\Assignment2> kubectl apply -f rs.yaml
replicaset.apps/frontend-rs created
PS C:\Users\hp\Documents\GenAI\Assignment2> kubectl apply -f deployment.yaml
deployment.apps/frontend-deployment created
PS C:\Users\hp\Documents\GenAI\Assignment2> kubectl apply -f service.yaml
service/frontend-service created
PS C:\Users\hp\Documents\GenAI\Assignment2> minikube service frontend-service
>>
```

Open the application and see output:

```
PS C:\Users\hp\Documents\GenAI\Assignment2> minikube service frontend-service
>>
```

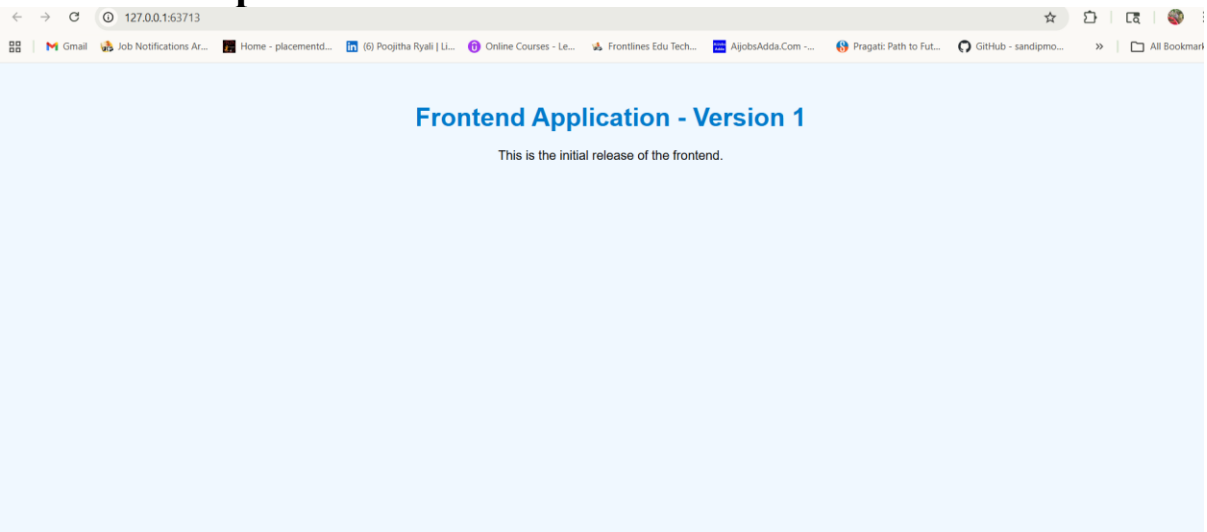
NAMESPACE	NAME	TARGET PORT	URL
default	frontend-service	80	http://192.168.49.2:30007

```
🔗 Starting tunnel for service frontend-service.
```

NAMESPACE	NAME	TARGET PORT	URL
default	frontend-service		http://127.0.0.1:63713

```
🎉 Opening service default/frontend-service in default browser...
❗ Because you are using a Docker driver on windows, the terminal needs to be open to run it.
✋ Stopping tunnel for service frontend-service.
```

Version 1 Output:

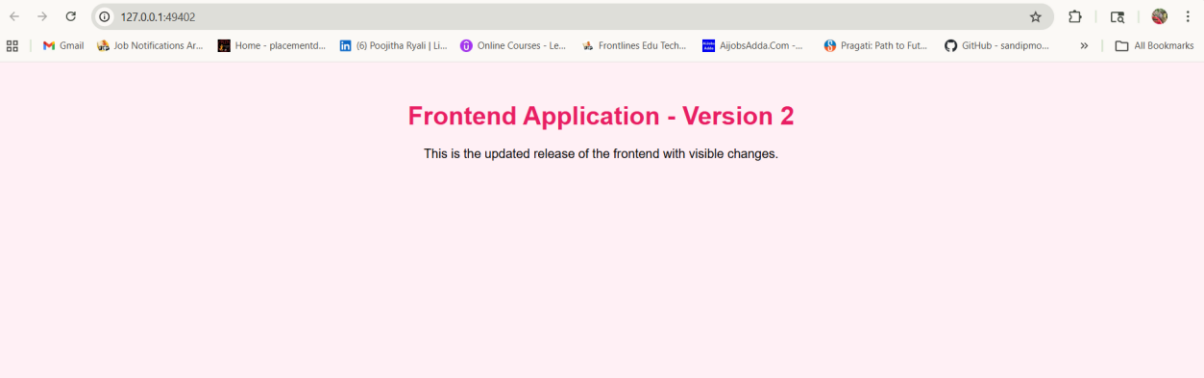


6. Rolling Update Demonstration

1. Deploy v1 → kubectl apply -f deployment.yml
2. Confirm Pods → kubectl get pods
3. Update image →

kubectl set image deployment/frontend-deployment frontend=rpoojitha04/frontend:v2
Observe rollout → kubectl rollout status deployment/frontend-deployment

Verify only v2 Pods remain.

```
PS C:\Users\hp\Documents\GenAI\Assignment2> kubectl set image deployment/frontend-deployment frontend=rpoojitha04/frontend:v2
deployment.apps/frontend-deployment image updated
PS C:\Users\hp\Documents\GenAI\Assignment2> kubectl rollout status deployment/frontend-deployment
Waiting for deployment "frontend-deployment" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "frontend-deployment" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "frontend-deployment" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "frontend-deployment" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "frontend-deployment" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "frontend-deployment" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "frontend-deployment" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "frontend-deployment" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "frontend-deployment" rollout to finish: 2 of 3 updated replicas are available...
Waiting for deployment "frontend-deployment" rollout to finish: 2 of 3 updated replicas are available...
deployment "frontend-deployment" successfully rolled out
```

Version 2 Output